

Class Test 2

Date : 1 November 2017
Time : 6:00 – 7:30 pm

Marks available: 52
Full marks: 50

No calculators may be used.

1. Read the following theorem and the outline of its proof:

Theorem. Suppose that a complex-valued function f is analytic on a region $D \subseteq \mathbb{C}$. If the set $Z = \{z \in \mathbb{C} : f(z) = 0\}$ has a limit point $z_1 \in D$, then $f(z) = 0$ for all $z \in D$.

Proof. Let L be the set of all limit points of Z . Note that L is a non-empty subset of Z . Consider any point $z_0 \in L$. Since D is open, there is an $R > 0$ such that the open disc $D_R = \{z \in \mathbb{C} : |z - z_0| < R\}$ is contained in D . It is possible to show that if $f(z) \neq 0$ for some $z \in D_R$, then $f(z) \neq 0$ for all z in a punctured disc about z_0 . This would contradict the fact that z_0 is a limit point of Z , so we must have that $f(z) = 0$ for all $z \in D_R$. It follows that $D_R \subseteq L$. Thus L is open. On the other hand, for each $z \in D \setminus L$, there is a punctured disc about z that does not contain any points from Z , so $D \setminus L$ is also open. Since D is connected, we conclude that $D \setminus L = \emptyset$. Thus $D \subseteq L \subseteq Z$. $\square$

Now answer the following questions:

- (a) Define a *limit point* of a set $A \subseteq \mathbb{C}$.
- (b) Why is L non-empty?
- (c) Why is L a subset of Z ?
- (d) In the proof, it is claimed that if $f(z) \neq 0$ for some $z \in D_R$, then $f(z) \neq 0$ for all z in a punctured disc about z_0 . Explain carefully why this is true. (HINT: Use the Taylor series expansion of f about z_0 .)
- (e) Why does the fact that $f(z) = 0$ for all $z \in D_R$ imply that $D_R \subseteq L$?
- (f) Why does the fact that D is connected allow us to conclude that $D \setminus L = \emptyset$?

[2, 1, 3, 6, 2, 3]

2. Does there exist a function f that is analytic on the open disc $D_1 = \{z \in \mathbb{C} : |z| < 1\}$ and that satisfies

$$f\left(\frac{1}{2k}\right) = f\left(\frac{1}{2k+1}\right) = \frac{1}{k}$$

for all positive integers k ?

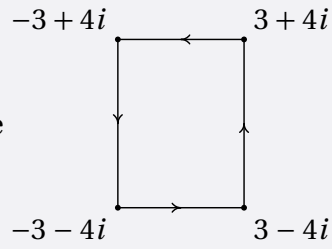
(A “Yes” or “No” answer won’t earn you any marks: you need to give a proper reason.)

[4]

3. Evaluate $\int_{\gamma} \frac{\exp(z^2)}{(z-2)^3} dz$, where $\gamma(t) = 3 + 2e^{it}$ for $0 \leq t \leq 2\pi$.

[5]

4. Evaluate $\int_{\mathcal{R}} \frac{6z^4}{32 - z^5} dz$, where $\mathcal{R}$ is the rectangle [6]



5. Use complex variables to evaluate $\int_0^\pi \frac{1}{1 + 8 \cos^2 \theta} d\theta$. [8]

6. Use the residue theorem to find P. V. $\int_{-\infty}^\infty \frac{x + 1}{x^2 + 4x + 5} dx$. [12]

Class Test 2 solutions

- 1.(a) $a \in \mathbb{C}$ is a *limit point* of A if every punctured disc centred at a contains a point from A .
- (b) Since $z_1 \in L$.
- (c) Each $z_0 \in L$ is the limit of a sequence of points in Z . Since $f(z) = 0$ for all $z \in Z$ and f is continuous at z_0 (because it's analytic on D), $f(z_0) = 0$.
- (d) Let $\sum_{k=0}^{\infty} a_k (z - z_0)^k$ be the Taylor series expansion of f about z_0 . This series converges to $f(z)$ on D_R . If $f(z) \neq 0$ for some $z \in D_R$, then at least one of the coefficients a_k is non-zero. Let n be the smallest non-negative integer with $a_n \neq 0$. Then
- $$f(z) = (z - z_0)^n g(z),$$
- where $g(z) = \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k$. Note that g is analytic on D_R and $g(z_0) = a_n \neq 0$. But this implies that $g(z) \neq 0$ for all z in an open disc centred at z_0 . Since $f(z) = (z - z_0)^n g(z)$, it follows that $f(z) \neq 0$ for all $z \neq z_0$ in that disc.
- (e) If $f(z) = 0$ for all $z \in D_R$, then each $z \in D_R$ is the centre of a disc contained in Z . This means that each $z \in D_R$ is a limit point of Z and is thus in L .
- (f) We have shown that both L and $D \setminus L$ are open. Since D is connected, this is only possible if L or $D \setminus L$ is empty. We have already noted that $L \neq \emptyset$, so $D \setminus L = \emptyset$.
2. Note that 0 is a limit point of both $A = \{1/(2k) : k \in \mathbb{N}\}$ and $B = \{1/(2k+1) : k \in \mathbb{N}\}$. On A we require $f(z) = 2z$, while on B we need $f(z) = 2z/(1-z)$. This contradicts the coincidence principle, so f can't be analytic on any set containing 0.
3. We can use Cauchy's formula for derivatives. Put $f(z) = \exp(z^2)$. Then

$$\int_{\gamma} \frac{f(z)}{(z-2)^3} dz = \frac{2\pi i}{2!} f''(2) = 18\pi i e^4.$$

Alternatively, you can use residues.

4. The integrand has five simple poles that all lie inside the rectangle $\mathcal{R}$. It's quicker to use the residue at infinity:

$$\begin{aligned} \int_{\mathcal{R}} \frac{6z^4}{32 - z^5} dz &= -2\pi i \operatorname{Res}_{z=\infty} \frac{6z^4}{32 - z^5} \\ &= 2\pi i \operatorname{Res}_{z=0} \frac{6}{z^6 (32 - z^5)} \\ &= 2\pi i \operatorname{Res}_{z=0} \frac{6}{z(32z^5 - 1)} \\ &= -12\pi i. \end{aligned}$$

5. Parametrise the unit circle by $\gamma = e^{i\theta}$, where $-\pi \leq \theta \leq \pi$. Since the integrand is an even function,

$$\begin{aligned} \int_0^\pi \frac{1}{1+8\cos^2\theta} d\theta &= \frac{1}{2} \int_{-\pi}^\pi \frac{1}{1+8\cos^2\theta} d\theta \\ &= \frac{1}{2} \int_{-\pi}^\pi \frac{1}{1+2(e^{i\theta} + e^{-i\theta})^2} d\theta \\ &= \frac{1}{2i} \int_{-\pi}^\pi \frac{e^{i\theta}}{2e^{4i\theta} + 5e^{2i\theta} + 2} \cdot ie^{i\theta} d\theta \\ &= \frac{1}{2i} \int_\gamma \frac{z}{2z^4 + 5z^2 + 2} dz \\ &= \frac{1}{2i} \int_\gamma \frac{z}{(2z^2 + 1)(z^2 + 2)} dz. \end{aligned}$$

The latter integrand is a rational function with simple poles at $\pm i/\sqrt{2}$ and at $\pm 2i$. Only the poles at $\pm i/\sqrt{2}$ are inside the unit circle. The residues at these two points are both

$$\frac{\pm i/\sqrt{2}}{(\pm 4i/\sqrt{2})(-1/2 + 2)} = \frac{1}{6}.$$

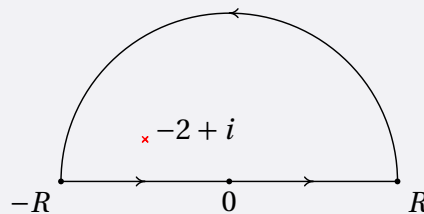
Thus

$$\int_0^\pi \frac{1}{1+8\cos^2\theta} d\theta = \frac{1}{2i} \cdot 2\pi i \cdot 2 \cdot \frac{1}{6} = \frac{\pi}{3}.$$

6. Note that

$$z^2 + 4z + 5 = 0 \iff z = -2 \pm i.$$

Let $\mathcal{C}$ be the curve made up of the segment of the real axis from $-R$ to R and the semicircle traced out by the path $\gamma(t) = Re^{it}$, where $0 \leq t \leq \pi$. We shall assume that $R > \sqrt{5}$, so the curve $\mathcal{C}$ encloses the pole at $-2 + i$.



Then

$$\int_{\mathcal{C}} = \int_{-R}^R + \int_\gamma.$$

The residue theorem says that

$$\int_{\mathcal{C}} \frac{z+1}{z^2+4z+5} dz = 2\pi i \cdot \operatorname{Res}_{z=-2+i} \frac{z+1}{z^2+4z+5} = 2\pi i \cdot \frac{(-2+i)+1}{2(-2+i)+4} = \pi(i-1).$$

To estimate the contribution from the semicircle, we use the parametrisation γ :

$$\begin{aligned}\int_{\gamma} \frac{z+1}{z^2+4z+5} dz &= \int_0^{\pi} \frac{Re^{it}+1}{R^2 e^{2it}+4Re^{it}+5} Rie^{it} dt \\ &= i \int_0^{\pi} \frac{e^{2it}+e^{it}/R}{e^{2it}+4e^{it}/R+5/R^2} dt \\ &\rightarrow i\pi \quad \text{as } R \rightarrow \infty.\end{aligned}$$

$$\text{Thus P. V. } \int_{-\infty}^{\infty} \frac{x+1}{x^2+4x+5} dx = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{x+1}{x^2+4x+5} dx = \pi(i-1) - i\pi = -\pi.$$